

(MR22-1SK0102) CAREER ADVANCEMENT SKILLS

COURSE OBJECTIVES:

By the end of this course, students will:

- Develop language and numerical skills to solve aptitude and reasoning problems.
- Enhance analytical thinking and problem-solving abilities.
- Develop version control, collaboration, and deployment skills for real-world projects.
- Enable to write optimized, modular, and error-free code using testing and debugging.
- Improve English communication, fluency, and corporate etiquette for professional readiness.

UNIT-I: Verbal & Quantitative Skills:

Verbal Ability: Nouns & Verbs, Prepositions, Tenses, Active & Passive Voice, Direct & Indirect Speech, Subject–Verb Agreement, Sentence Correction, Vocabulary, Sentence Completion, Para Jumbles, Reading Comprehension.

Quantitative Aptitude: Number Systems, LCM, HCF, Averages, Ratio & Proportion, Percentages, Problems on Ages, Profit & Loss, Discounts, Simple & Compound Interest, Time & Work, Pipes & Cisterns, Time, Speed & Distance, Trains, Boats, Permutations & Combinations.

UNIT-II: Logical Reasoning & Analytical Skills:

Clocks, Calendars, Directions, Set Theory & Logical Venn Diagrams, Syllogisms, Cubes & Dice, Seating Arrangement, Ranking & Ordering, Blood Relations, Series, Analogy, Odd Man Out, Coding & Decoding, Non-Verbal Reasoning, Coded Inequalities, Alphabet Test, Data Sufficiency.

UNIT-III: Git & Deployment:

Introduction to Git, Git Workflow (Working Directory, Staging Area, Repository), Core Git Commands, Branching & Collaboration, Deployment Basics, Automation & CI/CD.

UNIT-IV: Test-Case Driven Programming and Code Optimization:

Understanding Public vs Hidden Test Cases, Input Validation and Error Handling, Modular and Readable Code Practices, Test Environment Simulation with Timer and Auto-Evaluation, Debugging Techniques for Logical and Runtime Errors, Code Optimization for Time and Space Complexity.

UNIT-V: Versant & Corporate English:

Versant Test Structure and Scoring, Reading Sentences, Repeat Sentences, Answering Questions, Sentence Builds, Story Retelling, Open Questions, Pronunciation, Fluency, and Accent Neutralization, Business Email and Corporate Communication, Group Discussions and JAM Sessions, Mock Versant

REFERENCE BOOKS:

1. Sharma, Arun., & Upadhyay, M. (2016). *Verbal Ability and Reading Comprehension for the CAT*. Tata McGraw-Hill Education.
2. Aggarwal, R. S. (2021). *Quantitative Aptitude for Competitive Examinations* (29th ed.). S. Chand Publishing.
3. Aggarwal, R. S. (2021). *A Modern Approach to Logical Reasoning*. S. Chand Publishing.
4. Gandhi, R. (2020). *Head First Git: A Learner's Guide to Understanding Git*. O'Reilly Media.
5. Pearson Education. (n.d.). *Versant Test Preparation Guide*. Pearson English Assessment.

COURSE OUTCOMES:

- Demonstrates proficiency in solving verbal and quantitative problems accurately and efficiently.
- Exhibits strong analytical and logical reasoning skills for complex problem-solving.
- Capable of managing code, collaborating on GitHub, and deploying projects successfully.
- Develops high-quality, optimized programs while effectively debugging errors.
- Communicates professionally, participates confidently in discussions, and handles corporate correspondence effectively.